Samridhi Vibhanshu

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Education

Vellore Institute of Technology, Vellore Aug. 2023 – Present
B.Tech. in Computer Science and Engineering, Specialization in Business Studies *CGPA: 9.81/10*
Delhi Public School, Ranchi *Class XII (CBSE): 96.40%* Apr. 2023

Projects

Agentic Bug Repair | *Java 17, JDK Compiler API, Linux namespaces, JavaScript* GitHub | 2026
- Built an agent that repairs failing Java builds: indexes the repo into a **code graph**, ranks suspect methods from call-graph coverage, stack traces and hybrid **BM25/vector retrieval**, and tests candidate patches in parallel.
- Each patch runs in a **sandbox** (no network, resource limits, watchdog); a pull request is opened only if every test passes on a clean checkout. HMAC-verified GitHub webhook, bounded job queue, live dashboard; **148 tests**.

Order Service | *Java 17, java.util.concurrent, JDK HTTP server* GitHub | 2026
- E-commerce order backend over a JSON REST API that **never oversells**: per-SKU locks taken in sorted order prevent deadlock; 64 threads racing for the last 50 units always yield exactly 50 orders.
- **Idempotency keys** block double charges on retries; state machine + outbox; **30 tests** incl. 9,600 random workflows.

Mini-SQS and Route Planner | *Java 17, no dependencies* GitHub | GitHub (routes) | 2026
- **SQS-style queue**: visibility timeouts, DLQs, FIFO dedup, long polling; **26 tests** incl. 600 randomized sequences.
- Routing with Dijkstra, **A***, Held–Karp and 2-opt, checked against Floyd–Warshall and brute force (**19 tests**).

Risk-Constrained Emergency Control | *Python, Scikit-learn* GitHub | 2026
- Ventilation control that tries a small **reversible test action** first and escalates only once it is confirmed safe.
- Safety-gated classifier on 5,000 simulated records: **0.8% false-safe rate**, **92.9% harmful-class recall**.

Patent Applications (6 through the VIT IPR Cell; 5 published)

Autonomous Wildfire Prediction and Response System *App. No. 202641083188* Published, Jul. 2026
Predicts wildfire risk from weather and environmental sensor data and automatically triggers alerts and response actions.

Distributed Tremor Detection and Hazard Response in Mining *App. No. 202641075003* Published, Jun. 2026
Distributed IoT sensor nodes with on-device (edge) AI detect underground tremors early and raise hazard alerts.

Confidence-Weighted Opinion Lineage for Sentiment Aggregation *App. No. 202641115708* Published, Sep. 2026
Traces how opinions spread and down-weights copied or coordinated posts to recover genuine (organic) sentiment.

Predictive Trust Assessment and Audit Prioritization of IS Assets *App. No. 202641108535* Published, 2026
Models audit history as a temporal knowledge graph to forecast trust decay and rank IT assets for audit.

Risk-Constrained Emergency Control via Reversible Micro-Interventions Published, 2026
Tests a small, reversible control action first and escalates only once a safety gate confirms it is safe.

AI-Enabled Predictive Railway Collision Avoidance Filed, 2026
Network-aware collision-risk prediction that stays reliable under degraded sensor data and warns before conflicts.

Experience and Leadership

Computer Society of India (CSI-VIT) Apr. 2024 – Present
Board Member (Core Lead) | Full Stack Developer *Vellore, Tamil Nadu*
- Built **Riddler from scratch** (Node.js, MongoDB), CSI-VIT's international event platform with a **live leaderboard**, serving **1,000+ participants across 25+ countries**.
- Promoted to **Board Member (Core Lead)** in Sep. 2025; lead development, design, operations and outreach teams.
- Contributed to the **Laser Tag** platform (React.js, Tailwind CSS); built a GitHub contribution visualization.

Programme Representative, VIT *Student–faculty liaison for 200+ students* Jul. 2024 – Present

Achievements

- **Branch Topper**, B.Tech. CSE (Business Studies), VIT – since first year (CGPA 9.81/10).
- Mine tremor detection system **shortlisted for an NRDC grant**; currently under technology-readiness (TRL) review.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL, C, C++, R, HTML, CSS
Frameworks/Libraries: React.js, Redux, Tailwind CSS, Scikit-learn, Pandas, NumPy, pytest
Tools/Platforms: Git, GitHub Actions, Linux, Node.js, MongoDB, Postman, SQLite, OCI
Concepts: DSA, OOP, Concurrency, DBMS, REST APIs, Machine Learning, RAG, Edge AI, IoT
Certifications: OCI 2025 Certified Generative AI Professional; OCI 2025 Certified AI Foundations Associate